

Advanced (Carolina Reaper)

Q1. Simplify $\sqrt{64}$

- (A) 8
- (B) 4
- (C) 16
- (D) 2
- (E) 6

- (A) $4x$
- (B) $4\sqrt{x}$
- (C) $2x$
- (D) $2\sqrt{x}$
- (E) $8x$

Q2. Simplify $\sqrt[3]{27}$

- (A) 3
- (B) 9
- (C) 1
- (D) 6
- (E) 2

Q6. A clock strikes 12 times. If the time between strikes is 3 seconds and the first strike is at 12 : 00, what is the total time elapsed until the last strike in seconds?

- (A) 33
- (B) 36
- (C) 39
- (D) 30
- (E) 42

Q3. A carpenter has 48 feet of wood to build a treehouse. If she wants to make the sides of the treehouse a square with area 16 square feet, what is the length of each side in feet?

- (A) $\sqrt{16}$
- (B) $\sqrt{48}$
- (C) 4
- (D) 8
- (E) 2

Q7. A snail is at the bottom of a 20-foot well. Each day, it climbs 3 feet, but at night, it slips back 2 feet. How many days will it take for the snail to reach the top of the well?

- (A) 18
- (B) 20
- (C) 28
- (D) 25
- (E) 22

Q4. Simplify $x^{1/2} \cdot x^{1/4}$

- (A) $x^{3/4}$
- (B) $x^{1/4}$
- (C) $x^{2/3}$
- (D) $x^{1/2}$
- (E) $x^{3/2}$

Q8. Simplify $(64)^{1/3}$

- (A) 4
- (B) 16
- (C) 32
- (D) 2
- (E) 8

Q5. Simplify $\sqrt{16x^2}$

Open Ended Questions (FRQ)

Q9. A bakery is making a special cake with a layered square design, where each layer has an area of 64 square inches. If they want to add a border around the cake, increasing its area by 25%, what is the area of the larger square in square inches?

Q10. A person wants to save money for a new bike and decides to save 5 dollars per day for 30 days. If the person then decides to invest this money into a savings account that earns an annual interest rate of 2%, how much money will they have after 1 year, assuming the interest compounds annually?

Chef's Tips

- Tip 1: Check if students simplify radicals correctly
- Tip 2: Make sure students can convert between radical and rational exponent form
- Tip 3: Ensure students can apply radical concepts to real-world scenarios